



Ahmed Abudeeb

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ABOUT ME

I am a Master's student in Chemical Engineering at King Fahd University of Petroleum and Minerals (KFUPM), currently conducting thesis research focused on the development of a machine learning based equation of state (EoS). I hold a Bachelor's degree in Chemical Engineering from KFUPM, with a concentration in Renewable Energy and Energy Storage. My academic training includes advanced coursework in heat transfer, chemical kinetics, and thermodynamics. In addition to my technical background, I am highly effective in teamwork environments.

EDUCATION AND TRAINING

M.S. in Chemical Engineering

King Fahd University of Petroleum and Minerals (KFUPM) [2023 – 2026]

City: Dhahran | Country: Saudi Arabia | Website: <https://www.kfupm.edu.sa/Default.aspx>

Internship

King Fahd University of Petroleum and Minerals (KFUPM) [2022]

City: Dhahran | Country: Saudi Arabia | Website: <http://www.kfupm.edu.sa/Default.aspx>

B.S. in Chemical Engineering with Concentration in Renewable Energy and Energy Storage

King Fahd University of Petroleum and Minerals (KFUPM) [2018 – 2023]

City: Dhahran | Country: Saudi Arabia | Website: <https://www.kfupm.edu.sa/Default.aspx>

High School degree

High School degree at Prince Sultan High School [2015 – 2018]

City: Riyadh | Country: Saudi Arabia

WORK EXPERIENCE

 – Dhahran, Saudi Arabia

Teaching Assistant at KFUPM

[2023 – 2025]

- Graded assignments and examinations for **Reaction Kinetics** (undergraduate level course).
- Graded coursework for **Artificial Intelligence in Chemical Engineering** (graduate level course).

EXPERIENCE

[2024 – 2026]

Master's Thesis – Equation of State (EoS) Modeling Using Machine Learning

Conducting a research thesis focused on developing a novel equation of state (EoS) for Lennard-Jones fluids using machine learning. This work bridges chemical thermodynamics with modern AI algorithms for predictive modeling.

[2025]

Review on Metal Organic Frameworks (MOFs) for CO₂ Capture

Conducted a comprehensive review of the scientific literature examining recent advances in the application of metal organic frameworks (MOFs) for carbon dioxide capture. The study critically analyzed structure performance relationships, adsorption capacities, regeneration efficiencies, and scalability considerations, with an emphasis on identifying key challenges and opportunities for the practical deployment of MOF-based CO₂ capture technologies.

Link: <https://doi.org/10.1016/j.surfin.2025.107381>

[2024]

AI-Powered Kinetic Modeling of Naphtha Hydrotreatment Project

Collaborated in a team project to apply artificial intelligence for kinetic modeling of a commercial naphtha hydrotreating plant. The project involved modeling hydrodesulfurization kinetics using real operational data, preprocessing it for AI applications, and applying regression techniques to derive rate constants and catalyst activity functions.

[2023]

Senior Design Project

The handling of plastic waste presents an environmental issue, so the idea of the project was about an innovative solution to utilize the gasification process to recycle plastic waste into syngas. These syngas can then be converted into green methanol and excess Hydrogen from the process. Then using the excess Hydrogen in the Fuel Cell to Produce Renewable Energy.

[2022 – 2023]

Major Concentration in Renewable Energy and Energy Storage

My major concentration, which is considered as a minor, is Renewable Energy and Energy Storage. I took several courses in this concentration. In the first semester, I took the first course which is about introduction to energy markets, policies, and regulations. Also, I took another course about fuel cells, batteries, and supercapacitors. In the next semester, I took the third course which is about the fundamentals of renewable energy processes. In addition, I took another course about Solar Energy Conversion.

[2022]

Geothermal-Powered Hydrogen Production Process Project

This analysis considers a binary geothermal power plant. Geothermal water is used as the heat source in this study, and the working fluid passes through an actual closed thermodynamic cycle known as the organic Rankine cycle. Following that, the PEM electrolysis is supplied with electricity and heated water to drive the electrochemical reactions.

[2022]

Internship in Micro Activity Test Unit

Trained on the main method for assessing fluid catalytic cracking (FCC) catalysts in the lab, the micro activity test (MAT) unit. The training was for two months. The MAT Unit is a method for evaluating catalyst performance, which is also used to evaluate experimental and new commercial samples as well as samples taken from commercial operational units.

[2020]

Research about Hydroponics Technology to Ensure Food Security

Researched about Hydroponics method of growing plants without soil and the role of this method to ensure food security in future.

PUBLICATIONS

[2025]
CO₂ Capture using functionalized metal organic frameworks: A review This review provides a comprehensive overview of recent advances in MOFs for CO₂ capture, with a focus on how different functionalization strategies (amines, polymers, hydroxyl, carboxyl, graphene oxide) enhance adsorption performance and stability. We also discuss the impact of synthesis methods, process parameters, and computational studies on CO₂ uptake, and highlight the key challenges and future directions for developing scalable, high-performance sorbents.

LANGUAGE SKILLS

Mother tongue(s): Arabic

Other language(s):

English

LISTENING C1 READING C1 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Microsoft Office / Python programming language / Matlab

SKILL HIGHLIGHTS

Analytical skills

Great interpersonal skill

Report writing skills

Good in Teamwork